



Apple's next big thing

Tim Cook believes augmented reality is the next big thing in tech and the key to be at top. <http://dgit.in/TimCookAI>



Uber President quits

Jeff Jones joined just six months ago. Says firm's values are incompatible with his. <http://dgit.in/UberQuit>

From the labs



Yuka's skull and lower jaw on display in Japan in 2013

in temperatures. Grass, instead, does not absorb as much sunlight causing much of it to be reflected. These findings unravel how intimately the flora, fauna and climate of an ecosystem are linked and how introduction of a species like the Woolly Mammoth could be the just the element needed to sustain it.

have serious implications, ethically as well as physiologically. Mammoths, like elephants, are highly social animals and these new-age hybrids might lack social skills owing to the absence of peers. Critics remark that the entire project would reduce to creating a miniscule amount of hybrids to be used as show ponies for wealthy individuals. Then there is this annoying thing called chaos. Chaotic systems show a sensitive dependence on variables - small changes lead to massive deviations. On paper, the reanimation of a species and subsequent reintroduction into the environment might seem straightforward, but because the number of variables that the system depends on is so huge, a small variation could throw the entire ecosystem into disarray.

Motivation

The original idea of reviving creatures of the past, especially the Woolly Mammoth, is to the credit of a Russian scientist, Sergey Zimov. He envisioned the Pleistocene Park (similar to Jurassic Park but named after the Ice Age), an area where human intervention is restoring the habitat to the grassland like Mammoth Steppe, a form of cold climate African savanna. He believes that reanimating this ecosystem could potentially help prevent climate change. It was this radical idea, that convinced George Church (the leading scientist of this group at Harvard), to take up the revival of the mammoth.

Mammoth like creatures could help preventing the underground layers in tundra (permafrosts) from melting. Tundras are the highest source of carbon in the Arctic owing to their being a rich burial ground for a massive amount of flora and fauna of the past. If these melt, the sheer amount of methane released will exacerbate the damage done by carbon dioxide to the atmosphere by manifold. By punching through the snow, these huge herbivores let cold air pass over it, restoring these permafrosts to their original frozen state. During summer months when trees replace grass, they were responsible for knocking down trees, either when they sharpened their tusks/horns, indulged in a show of strength or just cleared their path. This helped in clearing the forests, which in case of the Arctic tundra is beneficial as the forest cover absorbs a lot of sunlight and heat leading to rise

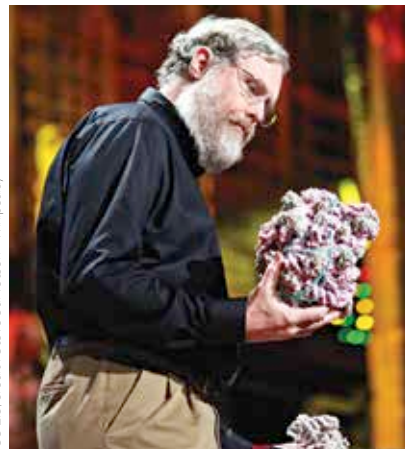
All the problems

Though resurrection biology seems to have caught the fancy of a lot people, not all are convinced by its tall claims. Conservation biologists believe that the governments around the world should try and preserve the present day fauna. Also, even if the mammoths return, owing to their limited source DNA (obtained from only few individuals), all the mammoths will end up have identical genetic makeup. Hence, mistakes in their DNA code will get accumulated over time leading to their ultimate wipeout. Additionally, the process of generating this hybrid utilises Asiatic elephant embryos, which is a nightmare for conservation biologists as these elephants are shrinking in terms of their numbers. On the other hand, nurturing an animal ex-vivo (outside a living being) would make the baby mammoth devoid of the essential contact it harbours with its mother right from its conception. This could

A resurrection - a revival

Like any new bold science project that catches the public attention, the Woolly Mammoth resurrection project surely has its pitfalls and has attracted its share of brickbats as well as having large scale benefits. Enthusiasts of the field believe that resurrection and re-animation (restoration of animals belonging to a region) are important as they garner attention and generate resources for conservation efforts as people might be interested in funding such "cool" programmes. Mammoths would bring back useful traits into the genetic pool, in turn helping the Asiatic elephant recover from the brink of extinction by making them more resistant to habitat change.

Woolly Mammoths were prey to a the fifth mass extinction event. We are now in midst of a sixth, triggered not by natural causes but by mankind. Any experiment that can reverse the damage done to Earth by humankind is worthy and no monetary based argument should be a valid one against it. Sometimes betting on the future of science can reap rich dividends. It worked for LIGO, for the LHC. We continue pumping money to the ITER project for nuclear fusion based energy. Trying to bring back a creature from the Ice Age as a biological experiment...Why not!



George Church - Harvard Biologist

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